

ECSE 2500
Lec 9
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Topic: Conditional PMF

Conditional Expected Values

Conditional Variance

Remember that the definition of Probability Mass function (PMF)

$$P_X(x_i) = P(X=x_i) = P(s \in S \mid X(s)=x_i)$$

↑
Sample space

We now combine this with the notion of conditional probability to talk about

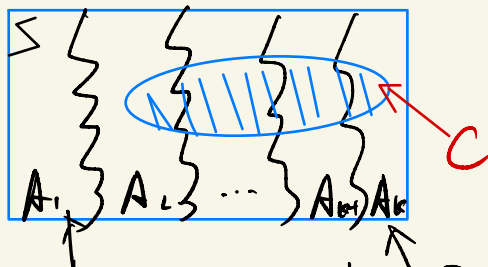
Conditional PMF

$$P_X(x_i | C) = P(X=x_i | C)$$

e.g., Probability that packet arrives in 5 seconds given that we have already waited for 5 seconds.

Alternatively, we can think of conditional PMF

$$P_X(x_i | C) = \frac{P(\{s \in S \mid X(s) = x_i \} \cap C)}{P(C)} \Leftarrow P(s \in C)$$



$$P(s \in S \mid X(s) = x_i) = P(A_i)$$

$$P(s \in S \mid X(s) = x_{i_c}) = P(A_{i_c})$$

Note: We implicitly assume that $P(C) > 0$ and $C \neq \emptyset$. Otherwise, $P_X(x_i | C) = 0$.

Example: What's the probability we got 5 heads on 5 biased coin tosses, given that the first two tosses were heads?

$X = \#$ of heads
in 5 coin tosses

$$P(\text{head}) = p, \quad P(\text{tail}) = 1 - p, \quad p \neq \frac{1}{2}.$$

$$P(X = 5 \mid \text{first 2 heads}) = \frac{P(\text{HHHHH} \cap \text{HHxxx})}{P(\text{HHxxx})}$$

$$= \frac{P(HHHHH)}{P(HHxxx)}$$

$$= \frac{P(H)^5}{P(H) \cdot P(H)} = p^3$$

If coin is fair, $p = \frac{1}{2}$,

$$P(X=5 | \text{first 2 heads}) = \frac{1}{8}$$

$$P(X=5) = \frac{1}{32} < < \frac{1}{8}$$

Notice: The notion $P_X(x_i | C)$ is still a probability mass function - satisfying all properties/rules of PMFs:

$$1) P_X(x_i | C) \geq 0 ;$$

$$2) \sum_{i=1}^N P_X(x_i | C) = 1, \text{ where } N \text{ is the number of possible values } X \text{ can take;}$$

3) Total probability theorem:

If $\{B_1, B_2, \dots, B_M\}$ is a partition of the original sample space, then

$$P_X(x_i) = \sum_{j=1}^M P_X(x_i | B_j) P(B_j)$$

Example: Waiting for a bus. Suppose that the

X : # of minutes we must wait is uniform,

$$P_X(k) = \begin{cases} \frac{1}{20} & , k = 1, 2, \dots, 20 \\ 0 & , \text{otherwise} \end{cases}$$

Given we have waited for 8 minutes, what is the conditional PMF of waiting time?

$$\begin{aligned} P_X(k | X > 8) &= \frac{P(X=k \text{ and } X > 8)}{P(X > 8)} \\ &= \frac{\frac{1}{20}}{\frac{12}{20}} \\ &= \frac{1}{12} \end{aligned}$$

Note: In the above calculation, we made the assumption that $k > 9$;

If $k \leq 8$, $P(X=k \text{ and } X > 8) = 0$,

$$P_X(k | X > 8) = 0$$

Example: Let

$X = \#$ of Additional Years a randomly chosen 70 years old person can live.

Event H : The person has high blood pressure;

X will be a Geometric RV with parameter $p=0.1$

Event N : The person has Normal blood pressure;

X will be a Geometric RV with parameter $p=0.05$.

Recall

$$P_X(k) = (1-p)^{k-1} p, \quad k=1, 2, \dots, \infty$$

$$P_{X|H}(k) = P_X(k|H) = (1-0.1)^{k-1} \cdot (0.1)$$

$$P_{X|N}(k) = P_X(k|N) = (1-0.05)^{k-1} \cdot (0.05)$$

If 40% of 70 years old person has high blood pressure, what is the PMF of X ?

$$\begin{aligned} P_X(k) &= P_{X|H}(k) \cdot P(H) + P_{X|N}(k) \cdot P(N) \\ &= (0.9)^{k-1} \cdot (0.1) \cdot (0.4) + (0.95)^{k-1} \cdot (0.05) \cdot (0.6) \end{aligned}$$

⇒ We can also define conditional expected values:

$$E[X|B] = \sum_{x_i \in S_X} x_i P_X(x_i|B)$$

If we know the conditional expected value of X , we can also compute

$$\begin{aligned} E[X] &= \sum_{x_i \in S_X} x_i P_X(x_i) \\ \text{Total probability theorem} &= \sum_{x_i \in S_X} x_i \sum_{j=1}^M P_X(x_i|B_j) P(B_j) \\ &= \sum_{j=1}^M \underbrace{\sum_{x_i \in S_X} x_i P_X(x_i|B_j)}_{E[X|B_j]} P(B_j) \\ &= \sum_{j=1}^M E[X|B_j] P(B_j) \end{aligned}$$

The above relation is convenient since we are often given the latter and we don't need to explicitly create the PMF.

Caveat : Variance does not follow the above relation in the sense that

$$\text{Var}(X) = \text{Var}(X|H) \cdot P(H) + \text{Var}(X|N) \cdot P(N)$$

What is conditional variance? How to calculate it?

$$\text{Var}(X|C) = E[X^2|C] - (E[X|C])^2$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

□ Memoryless property of Geometric Random Variable

Say X is a Geometric Random Variable with parameter p , i.e.,

$$P_X(k) = (1-p)^{k-1} \cdot p, \quad k=1, 2, \dots, \infty$$

What is $P(X > j+j_0 | X > j)$?

$$\rightarrow = P(X > j_0)$$

To show the above memoryless property,

$$\begin{aligned}P(X > j) &= \sum_{k=j+1}^{\infty} (1-p)^{k-1} \cdot p \\&= (1-p)^j \sum_{k=1}^{\infty} (1-p)^{k-1} \cdot p \\&= (1-p)^j\end{aligned}$$

Likewise, we can compute

$$\begin{aligned}P(X > j + j_0) &= \sum_{k=j+j_0+1}^{\infty} (1-p)^{k-1} \cdot p \\&= (1-p)^{j_0+j}\end{aligned}$$

Therefore, we have

$$\begin{aligned}P(X > j + j_0 | X > j) &= \frac{P(X > j + j_0)}{P(X > j)} \\&= \frac{(1-p)^{j_0+j}}{(1-p)^j} \\&= (1-p)^{j_0} = P(X > j_0)\end{aligned}$$

Interpretation of Memoryless Property:

If success hasn't occurred in the first j trials, the probability of rerunning j_0 more trials is the same as running j_0 trials in the first place

(i.e., coin does not "memorize" previous history if we fail, it is the same as starting again)